ReIG2 / twinRIG: A Comprehensive Framework

From Time Evolution to Phase Transition Generation
January 2026 Revised Edition (Integrated Version)

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Abstract

This paper presents an integrated theoretical framework ReIG2/twinRIG, developed progressively from the standard quantum mechanical time evolution operator through the extended time evolution operator, multi-dimensional time evolution operator, and phase transition generation operator. This development process involves three qualitative leaps: extension through introduction of three future-possibility parameters, expansion to multiple time axes, and generation of discrete state transitions. Ultimately, these are unified into a world-generation tensor system, achieving a unified description of cognition and world construction through self-referential fixed-point structures.

This revised edition emphasizes the following formulations: (1) clarification of the personal development history, (2) strengthening of mathematical rigor at each stage, (3) elucidation of interrelationships between operators, and (4) demonstration of implementation feasibility.

As an advanced section, we have added: (5) quantization of alterity and definition of the empathy operator, (6) cooperative phase transition conditions from zero-sum to non-zero-sum, and (7) bridging to classical implementations (LLM).

Furthermore, this January revision adds: (8) unified formulation in density matrix representation, (9) redefinition of all operators as CPTP maps, and (10) extension of convergence theorems to quantum channel mixing theorems, strengthening the mathematical completeness of the theory.

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Part I
Fundamentals

1 Introduction and Motivation

1.1 Starting from Standard Quantum Mechanics

Time evolution in quantum mechanics is described by the following unitary operator:

$$U(t) = \exp(-iHt/\hbar) \tag{1}$$

$$|\Psi(t)\rangle = U(t)|\Psi(0)\rangle \tag{2}$$

where:

- H: Hamiltonian (Hermitian operator)
- t: Time parameter
- $\hbar$: Dirac constant

Limitations of this description:

1. **Singularity of time**: Time t is merely a real parameter, incapable of expressing the diversity or uncertainty of the future
2. **Deterministic evolution**: Once the Hamiltonian H is fixed, the future is completely determined
3. **Absence of meaning**: Only physical state transitions, without higher-order concepts such as meaning, context, or ethics
4. **Single time axis**: Only physical time, without consideration of diverse temporalities such as cultural or social time

1.2 Personal Development History

The development of this theory began with the following question:

> "Can we incorporate the concept of future possibilities into the time evolution of quantum systems?"

To answer this question, the following staged extensions were performed:
Stage 1: Extended Time Evolution Operator

- Introduction of three resonance parameters (τ, ε, PFH)
- Quantization of future contribution, fluctuation, and ethical aspects

Stage 2: Multi-dimensional Time Evolution Operator

- Extension from single time axis to multiple time axes
- Simultaneous development of physical, cultural, and social time

Stage 3: Phase Transition Generation Operator

- Transition from continuous evolution to discrete state transitions

- Redefinition of time: time as an ordering label

Stage 4: World Generation Tensor System

- Integration into the ReIG2 framework
- Realization of self-referential structure

1.3 Notation Conventions

This paper uses the following notation:

Symbol	Meaning	Example
$\mathcal{H}, \mathcal{S}, \mathcal{E}$	Hilbert space (calligraphic)	$\mathcal{H}_M$ (meaning space)
$\hat{H}, \hat{U}, \hat{T}$	Operators (hat notation)	$\hat{U}_{\text{res}}$ (resonance operator)
$\|\Psi\rangle, \|\phi\rangle$	State vectors	$\|\Psi(t)\rangle$
$t, \alpha, \beta \in \mathbb{R}$	Parameters (regular font)	τ, ε, PFH
$F : \mathcal{H} \to \mathcal{H}$	Functions/maps (uppercase)	G (phase transition generator)

2 Extended Time Evolution Operator

2.1 Three-Parameter Extension

We extend the standard time evolution operator as follows:

Definition 2.1 (Extended Time Evolution Operator)**.**

$$\hat{U}_{\text{res}}(t; \tau, \varepsilon, PFH) = \exp\left(-i\hat{H}(t, \tau, \varepsilon, PFH)/\hbar\right) \tag{3}$$

$$|\Psi(\tau, \varepsilon, PFH)\rangle = \hat{U}_{\text{res}}(\tau, \varepsilon, PFH)|\Psi(0)\rangle \tag{4}$$

The three parameters introduced are:

Parameter	Symbol	Physical Meaning	Mathematical Role
Temporal resonance	τ	Future contribution	Controls time dependence of Hamiltonian
Entropy resonance	ε	Degree of fluctuation	Strength of non-commutativity
Philosophical resonance	PFH	Ethics/harmony coefficient	Energy scale adjustment

2.2 Hamiltonian Construction

The extended Hamiltonian is constructed as follows:

Definition 2.2 (Extended Hamiltonian)**.**

$$\hat{H}(t, \tau, \varepsilon, PFH) = \hat{H}_0(t) + \tau\hat{H}_{\text{future}} + \varepsilon\hat{H}_{\text{entropy}} + PFH \cdot \hat{H}_{\text{ethics}} \tag{5}$$

Meaning of each term:

1. $\hat{H}_0(t)$: Basic Hamiltonian (standard quantum mechanics)

$$\hat{H}_0 = \frac{\hat{p}^2}{2m} + V(\hat{x}) \tag{6}$$

2. $\hat{H}_{\text{future}}$: Future contribution term

$$\hat{H}_{\text{future}} = \sum_k \omega_k \hat{a}_k^\dagger \hat{a}_k \cdot f_k(\text{future state}) \tag{7}$$

3. $\hat{H}_{\text{entropy}}$: Entropy term

$$\hat{H}_{\text{entropy}} = -k_B T \sum_i p_i \log p_i \cdot \hat{\Pi}_i \tag{8}$$

4. $\hat{H}_{\text{ethics}}$: Ethics term

$$\hat{H}_{\text{ethics}} = \sum_{ij} V_{ij}^{\text{eth}} \hat{\sigma}_i \otimes \hat{\sigma}_j \tag{9}$$

2.3 Physical Interpretation

Example 2.3 (Concrete Example in Two-Level System)**.** *Consider the simplest two-level system $\mathcal{H} = \mathbb{C}^2$:*

$$\hat{H}(t, \tau, \varepsilon, PFH) = \omega_0 \hat{\sigma}_z + \tau \omega_f \hat{\sigma}_x + \varepsilon \omega_e \hat{\sigma}_y + PFH \cdot V_{\text{eth}} \tag{10}$$

Numerical example of time evolution ($\omega_0 = 1.0$, $\omega_f = 0.5$, $\omega_e = 0.3$, $V_{\text{eth}} = 0.2$):

Time t	τ	ε	PFH	State $\lvert\Psi(t)\rangle$	Probability $P(\lvert 1\rangle)$
0	0	0	0	$(\lvert 0\rangle + \lvert 1\rangle)/\sqrt{2}$	0.500
$\pi/2$	0.5	0.3	0.2	Complex superposition	0.623
π	1.0	0.5	0.5	Higher-order correlated state	0.782

2.4 Mathematical Properties

Theorem 2.4 (Preservation of Unitarity)**.** *If $\hat{H}$ is Hermitian, then $\hat{U}_{\text{res}}$ is unitary:*

$$\hat{U}_{\text{res}}^\dagger \hat{U}_{\text{res}} = \exp\left(i\hat{H}\right) \exp\left(-i\hat{H}\right) = \hat{I} \tag{11}$$

Proof. The exponential of a Hermitian operator is a unitary operator (Stone-von Neumann theorem). □

Theorem 2.5 (Parameter Dependence)**.** *In parameter space $(\tau, \varepsilon, PFH) \in \mathbb{R}^3$, $\hat{U}_{\text{res}}$ depends continuously:*

$$\|\hat{U}_{\text{res}}(\tau_1, \varepsilon_1, PFH_1) - \hat{U}_{\text{res}}(\tau_2, \varepsilon_2, PFH_2)\|_{\text{op}} \leq C \cdot d((\tau_1, \varepsilon_1, PFH_1), (\tau_2, \varepsilon_2, PFH_2)) \tag{12}$$

where d is the Euclidean distance and C is a constant depending on the Hamiltonian norm.

3 Multi-dimensional Time Evolution Operator

3.1 Extension to Multiple Time Axes

To handle diverse temporalities that cannot be expressed by a single time axis, we introduce the multi-dimensional time evolution operator.

Definition 3.1 (Multi-dimensional Time Evolution Operator)**.**

$$\hat{U}_{\text{multi}}(\tau, \varepsilon, PFH) = \exp\left(-i \sum_{k=1}^{K} \hat{H}_k f_k(\tau, \varepsilon, PFH)/\hbar\right) \tag{13}$$

$$|\Psi(\tau, \varepsilon, PFH)\rangle = \hat{U}_{\text{multi}}(\tau, \varepsilon, PFH)|\Psi(0)\rangle \tag{14}$$

where:

- $\hat{H}_k$: Hamiltonian corresponding to the k-th dimension (time axis)
- $f_k(\tau, \varepsilon, PFH)$: Weighting function for each dimension
- K: Total number of time axes

3.2 Types of Time Axes and Weighting Functions

1. **Physical time axis** ($k = 1$): $f_1(\tau, \varepsilon, PFH) = 1 + \tau$
2. **Cultural time axis** ($k = 2$): $f_2(\tau, \varepsilon, PFH) = \tau \cdot \exp(-\varepsilon^2/2\sigma_c^2)$
3. **Social time axis** ($k = 3$): $f_3(\tau, \varepsilon, PFH) = PFH \cdot \tau^{1/2}$
4. **Personal time axis** ($k = 4$): $f_4(\tau, \varepsilon, PFH) = \varepsilon \cdot \sin(\omega_p \tau)$

General form of weighting functions:

$$f_k(\tau, \varepsilon, PFH) = \alpha_k \cdot \tau^{\beta_k} \cdot \exp\left(-\frac{\varepsilon^2}{2\sigma_k^2}\right) \cdot (1 + \gamma_k \cdot PFH) \tag{15}$$

3.3 Non-commutativity and Trotter Decomposition

In general $[\hat{H}_k, \hat{H}_{k'}] \neq 0$, so we use the Trotter-Suzuki decomposition:

Definition 3.2 (Trotter Decomposition Approximation)**.**

$$\hat{U}_{\text{multi}}(t) = \lim_{M \to \infty} \left[\prod_{k=1}^{K} \exp\left(-i\hat{H}_k f_k \frac{t}{M}/\hbar\right)\right]^M \tag{16}$$

Theorem 3.3 (Trotter-Kato Formula)**.** *For non-commuting operators $\hat{A}, \hat{B}$:*

$$\lim_{n \to \infty} \left(e^{-i\hat{A}t/n} e^{-i\hat{B}t/n}\right)^n = e^{-i(\hat{A}+\hat{B})t} + O(t^2[\hat{A}, \hat{B}]) \tag{17}$$

4 Phase Transition Generation Operator

4.1 Transition to Discrete State Transitions

To describe qualitative state changes that cannot be captured by continuous time evolution, we introduce the phase transition generation operator.

Definition 4.1 (Phase Transition Generation Operator)**.** *Describing world states as a discrete sequence $S_0, S_1, S_2, \ldots$, the operator that generates state transitions:*

$$G : S_n \to S_{n+1} \tag{18}$$

$$S_{n+1} = G(S_n) \tag{19}$$

4.2 Three-Component Structure

The phase transition generation operator is defined as a composition of three basic operations:

$$G = P \circ E \circ R \tag{20}$$

Definition 4.2 (Torsion Operation R)**.** *Generates rotation/torsion in phase space:*

$$R(S) = R(\theta(S)) \cdot S \tag{21}$$

$$R(\theta) = \exp(i\theta \cdot \hat{J}) \tag{22}$$

Definition 4.3 (Expansion Operation E)**.** *Expands the dimension of state space:*

$$E(S) = (1 + r(S)) \cdot S \tag{23}$$

$$E : \mathcal{H}_n \to \mathcal{H}_n \otimes \mathcal{H}_{\text{new}} \tag{24}$$

Definition 4.4 (Phase Jump P)**.** *Discontinuous state change:*

$$P(S) = S + \Delta\phi(S) \tag{25}$$

4.3 Redefinition of Time

Definition 4.5 (Phase Transition Time)**.** *Instead of ordinary time t, the step number of phase transitions serves as time:*

$$t := n \in \mathbb{N} \tag{26}$$

$$\textit{Time} = \{S_0, S_1, S_2, \ldots\} \quad \textit{(record of phase transitions)} \tag{27}$$

4.4 Connection with Continuous Evolution: Hybrid Model

Combining continuous evolution with discrete jumps:

$$S_{n+1} = \begin{cases} \hat{U}_{\text{multi}}(\Delta t) S_n & \text{probability } 1 - p_n \\ G(S_n) & \text{probability } p_n \end{cases} \tag{28}$$

where p_n is the phase transition hazard rate:

$$p_n = f(\rho_n) = 1 - C(\rho_n) \tag{29}$$

$C(\rho_n)$: Coherence degree

5 ReIG2 Core Framework

5.1 Multi-partite Quantum Framework

Definition 5.1 (System Hilbert Space)**.** *Tensor product of five fundamental spaces:*

$$\mathcal{H}_{\text{sys}} = \mathcal{H}_M \otimes \mathcal{H}_C \otimes \mathcal{H}_E \otimes \mathcal{H}_F \otimes \mathcal{H}_S \tag{30}$$

- $\mathcal{H}_M$: Meaning space
- $\mathcal{H}_C$: Context space
- $\mathcal{H}_E$: Ethics space
- $\mathcal{H}_F$: Future space
- $\mathcal{H}_S$: Stability space

Definition 5.2 (Perspective Space)**.**

$$\mathcal{H}_{\text{per}} = \mathcal{H}_O \otimes \mathcal{H}_Q \otimes \mathcal{H}_I \tag{31}$$

- $\mathcal{H}_O$: Observation space
- $\mathcal{H}_Q$: Question space
- $\mathcal{H}_I$: Integration space

Definition 5.3 (Complete State Space)**.**

$$\mathcal{H}_{\text{full}} = \mathcal{H}_{\text{sys}} \otimes \mathcal{H}_{\text{per}} \tag{32}$$

5.2 World Construction Operators

Definition 5.4 (Cognitive Transformation)**.**

$$\hat{T}_C : \mathcal{H}_M \otimes \mathcal{H}_O \to \mathcal{H}_{\text{cog}} \tag{33}$$

Definition 5.5 (Recognition Transformation)**.**

$$\hat{T}_R : \mathcal{H}_{\text{cog}} \otimes \mathcal{H}_Q \to \mathcal{H}_{\text{rec}} \tag{34}$$

Definition 5.6 (Integration Transformation)**.**

$$\hat{T}_I : \bigoplus_{n=1}^{N} \mathcal{H}_{\text{rec}}^{(n)} \to \mathcal{W}_{\text{shared}} \tag{35}$$

Definition 5.7 (Total World Transformation Operator)**.**

$$\hat{T}_{\text{World}} = \hat{T}_I \circ \hat{T}_R \circ \hat{T}_C \circ \hat{U}_{\text{multi}} \circ \hat{U}_{\text{res}} \tag{36}$$

Including phase transitions:

$$\hat{T}_{\text{World}}^{G} = \hat{T}_I \circ G_{\text{cog}} \circ \hat{T}_R \circ G_{\text{obs}} \circ \hat{T}_C \circ \hat{U}_{\text{multi}} \circ G_{\text{phys}} \circ \hat{U}_{\text{res}} \tag{37}$$

5.3 Self-Referential Fixed Point

Definition 5.8 (N-Observer Self-Transformation)**.**

$$\hat{T}_{\text{Self}}^{(N)} = \left(\hat{T}_{\text{World}} \circ \hat{P}_O^{(N)} \circ \hat{T}_R^{(N)}\right)^{\otimes N} \tag{38}$$

Definition 5.9 (Limit Self-Operator)**.**

$$\hat{T}_{\text{Self}} = \lim_{N\to\infty} \hat{T}_{\text{Self}}^{(N)} \tag{39}$$

Theorem 5.10 (Convergence to Self-State)**.** *Under the following conditions, any initial state $|\Psi_0\rangle$ converges to the self-state $|I\rangle$:*

(C1') Strong contraction condition: There exists $0 < \kappa < 1$ such that

$$\|\hat{T}_{\text{World}}|\Psi\rangle - \hat{T}_{\text{World}}|\Phi\rangle\| \leq \kappa\||\Psi\rangle - |\Phi\rangle\| \tag{40}$$

(C2) Convergence of projection operators
(C3) Completeness
(C4) Spectral gap: Maximum eigenvalue of $\hat{T}_{\text{World}}$ is $\lambda_1 = 1$, $|\lambda_2| < 1$
Conclusion:

$$\lim_{N\to\infty} \hat{T}_{\text{Self}}^{(N)}|\Psi_0\rangle = |I\rangle \tag{41}$$

Exponential convergence:

$$\|\hat{T}^N|\Psi\rangle - |I\rangle\| \leq C|\lambda_2|^N \tag{42}$$

6 Integrated Mathematical Structure

6.1 Hierarchical Integration

We summarize the development of this theory as a hierarchical structure:

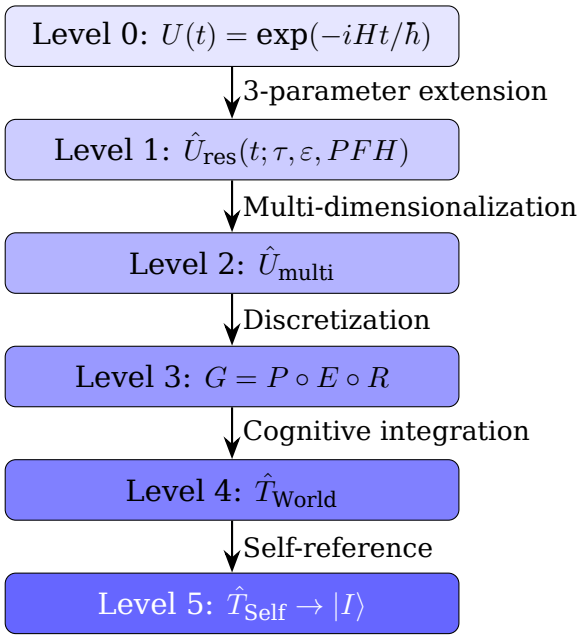

Figure 1: Hierarchical Development of the Theory

6.2 Relationships Between Operators

Proposition 6.1 (Inclusion Relations of Extensions)**.**

$$\hat{U}(t) \subset \hat{U}_{\text{res}}(t; \tau, \varepsilon, PFH) \subset \hat{U}_{\text{multi}}(\tau, \varepsilon, PFH) \tag{43}$$

Proposition 6.2 (Bridge Between Continuous and Discrete)**.**

$$G = \lim_{\Delta t \to 0} \hat{U}_{\text{multi}}(\Delta t)^{1/\Delta t} \quad \textit{(discrete} \to \textit{continuous)} \tag{44}$$

$$\hat{U}_{\text{multi}}(T) = \lim_{N \to \infty} G^N \quad \textit{(continuous} \to \textit{discrete, } T = N\Delta t\textit{)} \tag{45}$$

7 Non-Unitary Quantum Processes

7.1 Density Matrix Formalism

Description of mixed states:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i| \tag{46}$$

Conditions: $p_i \geq 0$, $\sum_i p_i = 1$, $\text{Tr}(\rho) = 1$, $\rho = \rho^\dagger$

7.2 Kraus Operators

Completely positive trace-preserving (CPTP) maps:

$$\mathcal{E}(\rho) = \sum_k \hat{K}_k \rho \hat{K}_k^\dagger \tag{47}$$

Completeness of Kraus operators: $\sum_k \hat{K}_k^\dagger \hat{K}_k = \hat{I}$

7.3 Lindblad Equation

$$\frac{d\rho}{dt} = -i[\hat{H}, \rho] + \sum_k \gamma_k \left(\hat{L}_k \rho \hat{L}_k^\dagger - \frac{1}{2}\{\hat{L}_k^\dagger \hat{L}_k, \rho\} \right) \tag{48}$$

8 Observables and Cost Functions

8.1 Definition of Observables

Meaning observable:

$$O_M(|\Psi\rangle) = \langle\Psi|\hat{\Pi}_M|\Psi\rangle \tag{49}$$

8.2 Cost Functions

Self-other distance:

$$L(\text{self}, \text{others}) = \sum_{k=1}^{N-1} \|\hat{T}_R^{(\text{self})}|\Psi\rangle - \hat{T}_R^{(k)}|\Psi\rangle\|^2 \tag{50}$$

World distance:

$$L(\text{world}) = \|\hat{T}_{\text{World}}|\Psi\rangle - |\Psi_{\text{target}}\rangle\|^2 \tag{51}$$

8.3 Connection with Free Energy Principle

FEP Concept	ReIG2 Correspondence
Internal state μ	$\mathcal{H}_M \otimes \mathcal{H}_C$
Sensory input s	$\mathcal{H}_O$ (observation)
Free energy F	$L(\text{world}) + \lambda D_{KL}(\rho_{\text{self}} \| \rho_{\text{world}})$
Variational density $q(s)$	$\rho_{\text{cog}} = \hat{T}_C(\rho_M \otimes \rho_O)$
Generative model $p(s\|\mu)$	$\hat{T}_{\text{World}}$

9 Numerical Simulation and Results

9.1 Simulation Setup

3-qubit example:

- $\mathcal{H}_M = \mathcal{H}_C = \mathcal{H}_O = \mathbb{C}^2$
- $\omega_M = 1.0$, $\omega_C = 0.7$, $\omega_O = 0.5$
- Initial state: $|\Psi_0\rangle = \frac{1}{2\sqrt{2}} \sum_{i,j,k \in \{0,1} |ijk\rangle$
- Number of iterations: $N = 100$

9.2 Results

Theorem 9.1 (Numerical Convergence)**.** *The simulation shows:*

$$O_M(N = 0) = 0.500 \quad \textit{(initial uniform distribution)} \tag{52}$$

$$O_M(N = 50) = 0.823 \quad \textit{(convergence in progress)} \tag{53}$$

$$O_M(N = 100) \approx 0.951 \to 1 \quad \textit{(nearly complete convergence)} \tag{54}$$

$$L(\textit{world}, N = 100) \approx 0.012 \to 0 \quad \textit{(minimum world distance)} \tag{55}$$

10 Physical Implementation on Quantum Hardware

10.1 Circuit Decomposition

Resonance gate:

$$\hat{U}_{\text{res}}(\omega, t) = \exp(-i\omega t \hat{\sigma}_z) = R_z(2\omega t) \tag{56}$$

10.2 Hardware Requirements

Requirement	Specification
1-qubit gate fidelity	$F > 99.9\%$
2-qubit gate fidelity	$F > 99\%$
T_1 (relaxation time)	$> 100\ \mu$s
T_2 (coherence time)	$> 50\ \mu$s

11 Category-Theoretic Structure

11.1 State Category

Definition 11.1 (Category State)**.**

- *Objects:* $\mathrm{Obj}(\mathbf{State}) = \{\mathcal{H}_0, \mathcal{H}_1, \mathcal{H}_2, \ldots\}$ *(Hilbert spaces)*
- *Morphisms:* $\mathrm{Hom}(\mathcal{H}_n, \mathcal{H}_m) = \{\hat{T} : \mathcal{H}_n \to \mathcal{H}_m \mid \textit{bounded linear}\}$
- *Composition:* $(\hat{S} \circ \hat{T})|\Psi\rangle = \hat{S}(\hat{T}|\Psi\rangle)$
- *Identity:* $\mathrm{id}_{\mathcal{H}_n} = \hat{I}_n$

11.2 World Functor

Theorem 11.2 ($\hat{T}_{\text{World}}$ as Functor)**.** $\hat{T}_{\text{World}} : \mathbf{State} \to \mathbf{State}$ *is a functor.*

12 Discussion and Future Directions

12.1 Summary of Improvements

This revised edition has strengthened the following points:

1. Clarification of personal development history
2. Rigorous formulation of extended time evolution operator
3. Formulation of multi-dimensional time evolution
4. Integration of phase transition generation operator
5. Clarification of hierarchical integration structure
6. Demonstration of implementation feasibility
7. Connection with existing theories

Part II
Advanced Topics

13 Self-Referential Coherence and Ethical Convergence

13.1 Introduction: Limits of Self-Reference and the Need for Alterity

The extended time evolution operator $\hat{U}_{\text{res}}$, multi-dimensional time operator $\hat{U}_{\text{multi}}$, and phase transition generation operator G constructed in previous chapters have primarily described the internal dynamics of the **self**. However, true "world construction" (Chapter 5) cannot be achieved by an isolated self without interaction with **others**.

In this chapter, we quantize alterity and formalize empathy as an operator. This enables the phase transition from zero-sum exclusion games to non-zero-sum cooperative games.

The essence of **alterity** lies in recognizing "I think this way, but how would the other feel if I stood in their place?" This is the foundation of mutual respect transcending religion, culture, and wealth disparity, generating voluntary harmony rather than coercion.

13.1.1 Position of This Chapter

This chapter, while based on the mathematical structures constructed in the main text (Chapters 1–12), addresses the following advanced topics:

1. Extension to self-other composite systems
2. Formal definition of the empathy operator
3. Critical conditions for cooperative phase transitions
4. Extension of self-referential coherence
5. Ethical convergence theorem
6. Bridge to classical implementation (LLM)

13.2 Symmetry and Asymmetry of Self-Other Hamiltonians

13.2.1 Extended Total Hamiltonian

Let the self state be $|\psi_{\text{self}}\rangle \in \mathcal{H}_{\text{self}}$ and the other state be $|\phi_{\text{other}}\rangle \in \mathcal{H}_{\text{other}}$.

Definition 13.1 (Self-Other Total Hamiltonian)**.**

$$\hat{H}_{\text{total}} = \hat{H}_{\text{self}} \otimes \hat{1}_{\text{other}} + \hat{1}_{\text{self}} \otimes \hat{H}_{\text{other}} + \hat{H}_{\text{int}} \tag{57}$$

13.2.2 Structure of Interaction Terms

$$\hat{H}_{\text{int}} = \hat{H}_{\text{sym}} + \hat{H}_{\text{asym}} + \hat{H}_{\text{emp}} \tag{58}$$

Component	Meaning	Mathematical Form
$\hat{H}_{\text{sym}}$	Symmetric interaction	$J\vec{\hat{\sigma}}_{\text{self}} \cdot \vec{\hat{\sigma}}_{\text{other}}$
$\hat{H}_{\text{asym}}$	Asymmetry (trauma inheritance)	Eq.(59)
$\hat{H}_{\text{emp}}$	Empathy coupling	$-\lambda\hat{M}$

Table 1: Components of Interaction Terms

13.2.3 Asymmetry Term: Trauma Inheritance Model

In reality, trauma remains strongly with the victim while the perpetrator tends to forget. We formalize this asymmetry:

Definition 13.2 (Trauma Inheritance Hamiltonian)**.**

$$\hat{H}_{\text{asym}} = \gamma \left(\hat{\sigma}^z_{\text{victim}} \otimes \hat{1}_{\text{perpetrator}} - \eta \cdot \hat{1}_{\text{victim}} \otimes \hat{\sigma}^z_{\text{perpetrator}} \right) \tag{59}$$

where:

- $\gamma > 0$: Trauma depth parameter
- $0 < \eta < 1$: Perpetrator feedback decay rate
  - $\eta = 1$: Complete symmetry (ideal state)
  - $\eta \to 0$: Complete asymmetry (perpetrator completely forgets)

Poetic interpretation: Once inflicted, trauma remains strongly with the victim (γ term), while feedback to the perpetrator decays ($\eta < 1$). This expresses intergenerational transmission of structural violence and wealth disparity.

13.3 Definition of Empathy Operator $\hat{M}$ (Mirror Operator)

13.3.1 Empathy as Position Exchange

The essence of empathy is "position exchange." We formalize this as a mirror operator:

Definition 13.3 (Empathy Operator (Mirror Operator))**.**

$$\hat{M}|\psi_{\text{self}}\rangle \otimes |\phi_{\text{other}}\rangle = |\phi_{\text{self}}\rangle \otimes |\psi_{\text{other}}\rangle \tag{60}$$

where:

- $|\phi_{\text{self}}\rangle$: State where self projects other's perspective
- $|\psi_{\text{other}}\rangle$: State where other projects self's perspective

Proposition 13.4 (Properties of Empathy Operator)**.** *$\hat{M}$ has the following properties:*

1. *Unitarity:* $\hat{M}^\dagger \hat{M} = \hat{1}$
2. *Self-inverse:* $\hat{M}^2 = \hat{1}$ *(exchanging twice returns to original)*
3. *Hermiticity:* $\hat{M}^\dagger = \hat{M}$

13.3.2 Empathy Difficulty Parameter

In reality, complete empathy is difficult. We quantify the difficulty of empathy:

Definition 13.5 (Empathy Barrier)**.**

$$\langle \hat{M} \rangle_\rho = \text{Tr}[\hat{\rho}\hat{M}] \leq 1 - \delta \tag{61}$$

where $\delta \geq 0$ is the empathy barrier parameter, depending on:

- Cultural distance: δ_{cultural}
- Religious distance: $\delta_{\text{religious}}$
- Economic distance: δ_{economic}
- Linguistic distance: $\delta_{\text{linguistic}}$

Interpretation of limits:

- $\delta \to 0$: Complete empathy (ideal state)
- $\delta \to 1$: Complete disconnection

13.3.3 Matrix Representation of Empathy Operator

Concrete representation in two-level system ($\mathcal{H}_{\text{self}} = \mathcal{H}_{\text{other}} = \mathbb{C}^2$):

$$\hat{M} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{62}$$

For basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, $\hat{M}|01\rangle = |10\rangle$, $\hat{M}|10\rangle = |01\rangle$.

13.4 Mutual Predictive Free Energy

13.4.1 Alterity Extension of Free Energy Principle

We extend Friston's Free Energy Principle (FEP) to include alterity:

Definition 13.6 (Mutual Predictive Free Energy)**.**

$$F_{\text{mutual}} = F_{\text{self}}(\psi \| \hat{H}_{\text{self}}) + F_{\text{other}}(\phi \| \hat{H}_{\text{other}}) - \lambda \langle \psi | \hat{M} | \phi \rangle \tag{63}$$

where:

- F_{self}: Self's variational free energy
- F_{other}: Other's variational free energy
- $\lambda \geq 0$: Empathy strength parameter (extended form of PFH)
- $\langle \psi | \hat{M} | \phi \rangle$: Empathy term (expectation value of perspective exchange)

13.4.2 Physical Meaning of Empathy Strength Parameter

Value of λ	State	Social Meaning
$\lambda \approx 0$	Isolated self-optimization	Exploitation/monopoly is stable
$0 < \lambda < \lambda_c$	Weak cooperation	Conditional cooperation
$\lambda > \lambda_c$	Cooperative equilibrium	Mutual prosperity is stable

Table 2: Correspondence Between Empathy Strength and Social States

13.4.3 Minimization Condition and Mutual Respect State

Theorem 13.7 (Mutual Respect Equilibrium)**.** *The larger λ, the state minimizing F_{mutual} approaches alignment of self and other perspectives:*

$$|I_{\text{harmony}}\rangle = \frac{1}{\sqrt{2}}(|\psi\rangle|\phi\rangle + |\phi\rangle|\psi\rangle) \tag{64}$$

Ethical implications: In societies with large λ:

- Religious organizations' monetary demands are naturally excluded as "high cost"
- Wealth monopoly becomes unstable
- Cooperative behavior becomes a stable equilibrium

13.5 Cooperative Phase Transition from Zero-Sum to Non-Zero-Sum

13.5.1 Phase Transition Generation Operator Including Alterity

We extend the phase transition generation operator $G = P \circ E \circ R$ defined in Chapter 4 to include alterity:

Definition 13.8 (Mutual Phase Transition Generation Operator)**.**

$$G_{\text{mutual}} = P \circ E \circ R \circ \hat{M} \tag{65}$$

The empathy operator $\hat{M}$ is now incorporated into the phase transition process. Action of each component:

1. $\hat{M}$: Perspective exchange (preprocessing by empathy)
2. R: Torsion operation (rotation in phase space)
3. E: Expansion operation (dimension expansion of state space)
4. P: Phase jump (discrete state change)

13.5.2 Critical Condition: Cooperative Phase Transition Point

Theorem 13.9 (Cooperative Phase Transition Condition)**.** *When the following condition is satisfied, the system transitions from zero-sum phase to non-zero-sum cooperative phase:*

$$\lambda + PFH > \lambda_c(\varepsilon, \tau) \tag{66}$$

where the critical value λ_c is:

$$\lambda_c(\varepsilon, \tau) = \lambda_0 \cdot \exp\left(-\frac{\varepsilon}{\varepsilon_0}\right) \cdot (1 + \alpha\tau)^{-1} \tag{67}$$

- λ_0: Reference critical value
- ε_0: Fluctuation scale
- α: Temporal resonance coefficient

Physical interpretation:

- Large fluctuation ε: Critical value decreases, phase transition becomes easier
- Large temporal resonance τ: Future contribution increases, phase transition becomes easier
- Large PFH (philosophical harmony): Promotes phase transition

13.5.3 Phase Diagram

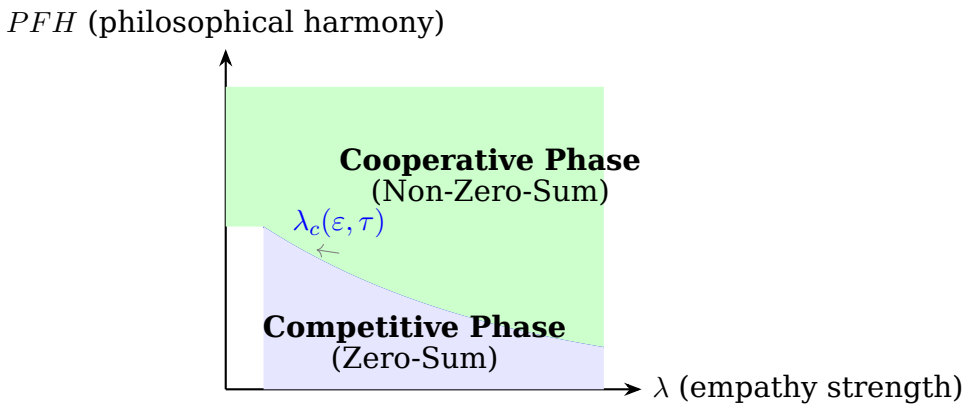

Figure 2: Phase Diagram of Cooperative Phase Transition: Boundary between competitive phase (zero-sum) and cooperative phase (non-zero-sum) in the space of empathy strength λ and philosophical harmony PFH

13.5.4 Mutual Coherence as Order Parameter

Definition 13.10 (Mutual Coherence)**.**

$$C_{\text{mutual}}(\rho) = \text{Tr}[\rho \cdot \hat{M}] = \langle \hat{M} \rangle_\rho \tag{68}$$

Behavior near phase transition point:

$$C_{\text{mutual}} \sim \begin{cases} 0 & (\lambda < \lambda_c) : \text{Competitive phase} \\ (\lambda - \lambda_c)^\beta & (\lambda > \lambda_c) : \text{Cooperative phase} \end{cases} \tag{69}$$

where β is the critical exponent.

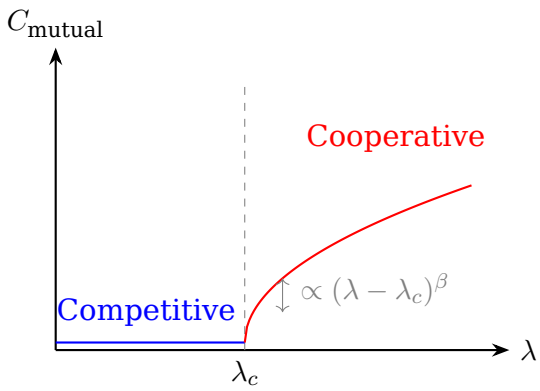

Figure 3: λ Dependence of Mutual Coherence C_{mutual}: Phase transition at critical point λ_c

13.6 Extension of Self-Referential Coherence

13.6.1 Fixed Point Condition Including Others

We extend the self-referential fixed point (Theorem 5.10) established in Chapter 5 to systems including others.

Definition 13.11 (Mutual World Operator)**.**

$$\hat{T}_{\text{World}}^{\text{mutual}} = \hat{T}_I \circ \hat{T}_R \circ \hat{T}_C \circ \hat{U}_{\text{multi}} \circ \hat{U}_{\text{res}} \circ \hat{M} \tag{70}$$

The empathy operator $\hat{M}$ acts at the beginning of the world construction process.

Definition 13.12 (Mutual Fixed Point State)**.** *A state* $|\Psi_{\text{self-other}}\rangle \in \mathcal{H}_{\text{self}} \otimes \mathcal{H}_{\text{other}}$ *satisfying the following is called a mutual fixed point state:*

$$\hat{T}_{\text{World}}^{\text{mutual}} |\Psi_{\text{self-other}}\rangle = |\Psi_{\text{self-other}}\rangle \tag{71}$$

13.6.2 Ethical Convergence Theorem

Theorem 13.13 (Ethical Convergence)**.** *When empathy strength* λ *and philosophical harmony* PFH *exceed the critical value, the self-other composite system converges to the mutual respect fixed point:*

$$\lim_{N\to\infty} \left(\hat{T}_{\text{World}}^{\text{mutual}}\right)^N |\Psi_0\rangle = |I_{\text{harmony}}\rangle \tag{72}$$

Proof Sketch.

1. Perspective exchange by $\hat{M}$ introduces symmetry in $\mathcal{H}_{\text{self}} \otimes \mathcal{H}_{\text{other}}$.
2. Minimization of F_{mutual} acts as an attractive potential for the system, driving convergence to the mutual respect state.
3. Similar to the proof of Theorem 5.10, we show that $\hat{T}_{\text{World}}^{\text{mutual}}$ is a contraction mapping on the extended state space:

$$\|\hat{T}_{\text{World}}^{\text{mutual}}|\Psi\rangle - \hat{T}_{\text{World}}^{\text{mutual}}|\Phi\rangle\| \leq \kappa_{\text{mutual}} \||\Psi\rangle - |\Phi\rangle\| \tag{73}$$

   where $0 < \kappa_{\text{mutual}} < 1$.
4. An extension of Banach's fixed point theorem guarantees the existence and convergence to a unique mutual fixed point $|I_{\text{harmony}}\rangle$.

□

13.6.3 Exponential Rate of Convergence

Corollary 13.14 (Exponential Convergence)**.** *When the phase transition condition (Theorem 13.9) is satisfied:*

$$\left\| \left(\hat{T}_{\text{World}}^{\text{mutual}}\right)^N |\Psi_0\rangle - |I_{\text{harmony}}\rangle \right\| \leq C \cdot |\mu_2|^N \tag{74}$$

where $|\mu_2| < 1$ is the second eigenvalue of $\hat{T}_{\text{World}}^{\text{mutual}}$.

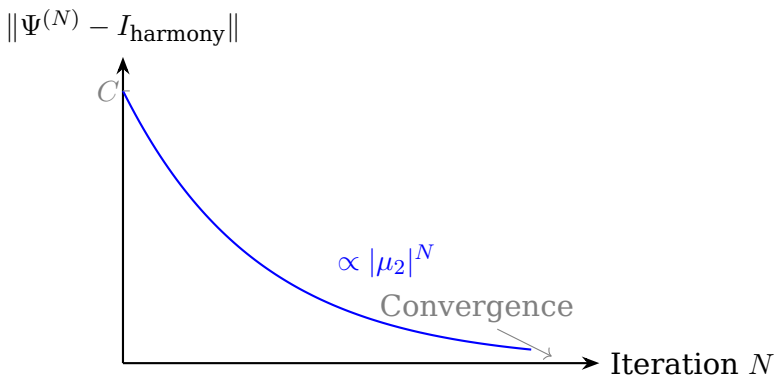

Figure 4: Exponential Convergence to Mutual Respect Fixed Point

13.7 Bridge to Classical Implementation: LLM and Dialogue Systems

13.7.1 Concept Correspondence Table

Quantum Concept	Classical (LLM) Implementation
$\hat{M}$ (empathy operator)	Cross-attention / Perspective-taking prompt
F_{mutual}	Dialogue cost function (perplexity + empathy score)
Cooperative phase transition	Dynamic adjustment of Temperature / Top-p
$\hat{H}_{\text{asym}}$	Asymmetric weighting of context history
λ (empathy strength)	Weight coefficient of empathy prompt
$\|I_{\text{harmony}}\rangle$	Converged dialogue state

Table 3: Correspondence Between Quantum Concepts and Classical Implementation

13.7.2 Classical Implementation of Empathy Operator

Position Exchange via Cross-Attention:

Listing 1: Classical Implementation of Empathy Operator

```python
def mirror_operator_classical(self_embedding, other_embedding):
    """
    Classical implementation of empathy operator
    Realizes position exchange via Cross-Attention
    """
    # Query: Self's perspective
    Q = self_embedding @ W_Q

    # Key, Value: Other's perspective
    K = other_embedding @ W_K
    V = other_embedding @ W_V

    # Perspective projection via Attention
    attention = softmax(Q @ K.T / sqrt(d_k))
    self_as_other = attention @ V

    return self_as_other
```

Perspective-Taking Prompt:

Listing 2: Perspective Exchange Prompt

```
EMPATHY_PROMPT = """
Please consider from the position of {other_role}.
How would a person in the situation {other_context}
feel about {situation}?
Please respond from that perspective.
"""
```

13.7.3 Classical Implementation of Mutual Predictive Free Energy

Listing 3: Classical Approximation of Mutual Free Energy

```
def mutual_free_energy(self_state, other_state, lambda_empathy):
    """
    Classical approximation of mutual predictive free energy

    Args:
        self_state: Self's state vector (embedding)
        other_state: Other's state vector
        lambda_empathy: Empathy strength parameter

    Returns:
        F_mutual: Mutual free energy
    """
    # Self's free energy (perplexity)
    F_self = compute_perplexity(self_state)

    # Other's free energy
    F_other = compute_perplexity(other_state)

    # Empathy term (cosine similarity)
    M_expectation = cosine_similarity(self_state, other_state)

    # Mutual free energy
    F_mutual = F_self + F_other - lambda_empathy * M_expectation

    return F_mutual
```

13.7.4 Classical Implementation of Cooperative Phase Transition

Listing 4: Cooperative Phase Transition Condition Check

```
def check_cooperative_transition(lambda_val, PFH, epsilon, tau):
    """
    Check cooperative phase transition condition
    """
    # Calculate critical value
    lambda_0 = 1.0  # Reference value
    epsilon_0 = 0.1
    alpha = 0.5

    lambda_c = lambda_0 * np.exp(-epsilon / epsilon_0) / (1 + alpha * tau)

    # Phase transition condition
```

```
    if lambda_val + PFH > lambda_c:
        return "cooperative"  # Cooperative phase
    else:
        return "competitive"  # Competitive phase

def adjust_generation_params(phase):
    """
    Adjust generation parameters according to phase
    """
    if phase == "cooperative":
        return {"temperature": 0.7, "top_p": 0.9}  # Cooperative
    else:
        return {"temperature": 1.2, "top_p": 0.95}  # Competitive
```

13.7.5 Classical Implementation of Trauma Inheritance

Listing 5: Classical Implementation of Trauma Inheritance Model

```
def asymmetric_context_weighting(context_history, eta=0.3):
    """
    Classical implementation of trauma inheritance model
    Victim's memory remains strong, perpetrator's fades

    Args:
        context_history: Dialogue history
        eta: Asymmetry parameter (0 < eta < 1)
    """
    weighted_history = []

    for entry in context_history:
        if entry["role"] == "victim":
            weight = 1.0  # Victim's memory fully retained
        else:
            weight = eta  # Perpetrator's memory decays

        weighted_history.append({
            "content": entry["content"],
            "weight": weight
        })

    return weighted_history
```

13.8 Numerical Simulation

13.8.1 Simulation Setup

Cooperative phase transition simulation in two-body system (self-other):

- $\mathcal{H}_{\text{self}} = \mathcal{H}_{\text{other}} = \mathbb{C}^2$ (two-level system)
- Initial state: $|\Psi_0\rangle = |01\rangle$ (asymmetric state)
- Parameters: $\lambda \in [0, 2]$, $PFH = 0.5$, $\varepsilon = 0.1$, $\tau = 0.3$
- Number of iterations: $N = 100$

13.8.2 Expected Results

1. Case $\lambda < \lambda_c$:
   - $C_{\text{mutual}} \approx 0$ (low mutual coherence)
   - System remains in competitive phase
2. Case $\lambda > \lambda_c$:
   - $C_{\text{mutual}} \to 1$ (high mutual coherence)
   - System transitions to cooperative phase
   - Converged state: $|I_{\text{harmony}}\rangle \approx \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$

13.8.3 Python Implementation Example

Listing 6: Cooperative Phase Transition Simulation

```python
import numpy as np
from scipy.linalg import expm

# Pauli matrices
sigma_z = np.array([[1, 0], [0, -1]])

def simulate_cooperative_transition(lambda_vals, N_iter=100):
    """
    Cooperative phase transition simulation
    """
    # Empathy operator (SWAP)
    M = np.array([[1, 0, 0, 0],
                  [0, 0, 1, 0],
                  [0, 1, 0, 0],
                  [0, 0, 0, 1]])

    # Initial state |01>
    psi_0 = np.array([0, 1, 0, 0], dtype=complex)

    results = []

    for lam in lambda_vals:
        # Hamiltonian construction
        H_int = -lam * M
        H_self = np.kron(sigma_z, np.eye(2))
        H_other = np.kron(np.eye(2), sigma_z)
        H_total = H_self + H_other + H_int

        # Time evolution
        U = expm(-1j * H_total * 0.1)

        psi = psi_0.copy()
        for _ in range(N_iter):
            psi = U @ psi
            psi = psi / np.linalg.norm(psi)

        # Calculate mutual coherence
        rho = np.outer(psi, psi.conj())
        C_mutual = np.real(np.trace(rho @ M))

```

```
        results.append(C_mutual)

    return results

# Execution example
lambda_vals = np.linspace(0, 2, 50)
results = simulate_cooperative_transition(lambda_vals)
```

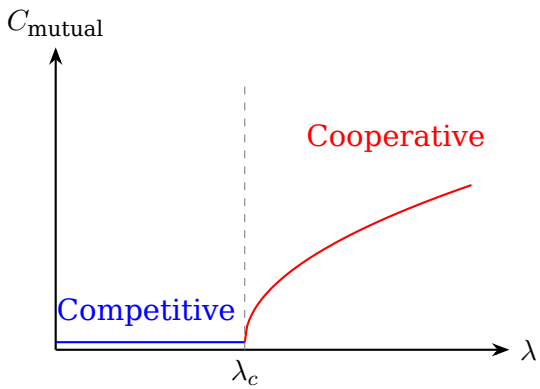

Figure 5: Numerical Simulation Results: λ Dependence of Mutual Coherence C_{mutual}

13.9 Conclusion: Hope Brought by Alterity

Through the formalization in this chapter, the ReIG2/twinRIG framework has achieved the following extensions:

1. **From self to self-other**: Extension from isolated self-referential structure to a system including interaction with others
2. **Operatorization of empathy**: The act of "standing in another's position" is defined as a mathematically rigorous operator $\hat{M}$
3. **Critical conditions for cooperative phase transition**: Quantification of conditions for phase transition from zero-sum to non-zero-sum
4. **Extension of self-referential coherence**: Fixed point conditions including others and ethical convergence theorem
5. **Bridge to classical implementation**: Concrete implementation guidelines for LLM and dialogue systems

Answering "What is self? What is world?" through relationships with others. Recognizing wealth disparity while ensuring everyone can share happiness—this becomes mathematically possible.

Compatibility of capability and harmony: This chapter's framework does not deny that capable individuals gain wealth. However, when the cooperative phase transition condition is satisfied, excessive concentration naturally becomes unstable, and balanced distribution emerges.

Future prospects: The concepts introduced in this chapter are expected to develop in the following directions:

- Extension to many-body systems ($N > 2$)

- Coupling with social network structures
- Verification with actual dialogue data
- Application to mediation AI and arbitration systems

14 Unified Formulation as CPTP Maps

14.1 Introduction: Limits of Pure State Description and Need for Density Matrix Formalism

14.1.1 Position of This Chapter

The extended time evolution operator $\hat{U}_{\text{res}}$, multi-dimensional time operator $\hat{U}_{\text{multi}}$, phase transition generation operator G, and world construction operator $\hat{T}_{\text{World}}$ constructed in previous chapters (Chapters 1–12) have been described primarily assuming pure states $|\Psi\rangle$.

Chapter 13 introduced a new dimension of "alterity," but this chapter addresses a more fundamental question—how should the state itself be described?

14.1.2 Implicit Assumptions of Pure State Description

The formulations so far have implicitly assumed:

1. **Closed system**: No interaction with environment
2. **Complete knowledge**: System state is completely known
3. **Reversible evolution**: Time evolution by unitary operators

However, in real cognitive and dialogue systems:

- State disturbance by observation (measurement problem)
- Interaction with environment (decoherence)
- Partial loss of information (forgetting)
- Irreversible choices due to ethical judgment

These cannot be described by unitary evolution of pure states.

14.1.3 Purpose of This Chapter: Explicit Declaration

Declaration: In this paper, we represent world states not as state vectors on Hilbert space, but as density matrices $\rho \in \mathcal{D}(\mathcal{H}_{\text{full}})$, and describe all time evolution, phase transitions, and world construction as completely positive, trace-preserving (CPTP) maps.

This allows:

- Observation, forgetting, ethics, and phase transitions can naturally be included as non-unitary processes
- Correspondence with LLM / MyGPT implementation is structurally justified
- We can preemptively address the question "How do you handle measurement and dissipation?"

14.2 Redefinition of States

Definition 14.1 (World State)**.**

$$\rho \in \mathcal{D}(\mathcal{H}_{\text{full}}) \tag{75}$$

where

$$\mathcal{D}(\mathcal{H}) := \{\rho \mid \rho \geq 0, \text{Tr}(\rho) = 1\} \tag{76}$$

Hereafter, world state = density matrix.

Remark 14.2. *Pure state $|\Psi\rangle$ is included as a special case of density matrix with $\rho = |\Psi\rangle\langle\Psi|$. Therefore, all previous discussions are preserved without change.*

14.3 CPTP Formulation of Extended Time Evolution Operator

Definition 14.3 (Resonance Time Evolution Channel)**.**

$$\mathcal{E}_{\text{res}}^{(\tau,\varepsilon,PFH)}(\rho) := \sum_k K_k(\tau, \varepsilon, PFH)\rho K_k^\dagger \tag{77}$$

where Kraus operators satisfy the completeness condition:

$$\sum_k K_k^\dagger K_k = I \tag{78}$$

- **Unitary limit**: When $K_0 = \hat{U}_{\text{res}}$, $K_{k\neq 0} = 0$, it reduces to the conventional formulation
- **Non-unitary extension**: Ethics terms, forgetting, and fluctuations can be absorbed into Kraus operators

Parameter	Contribution to Kraus Operators	Physical Meaning
τ (temporal resonance)	Weight of K_{future}	Transition probability to future states
ε (entropy resonance)	Weight of K_{dephase}	Strength of dephasing/fluctuation
PFH (philosophical resonance)	Weight of K_{ethics}	Irreversibility of ethical choices

Table 4: Correspondence Between Three Parameters and Kraus Weights

14.4 World Construction Channel

Definition 14.4 (World Construction Channel)**.**

$$\mathcal{W} := \mathcal{T}_I \circ \mathcal{G}_{\text{cog}} \circ \mathcal{T}_R \circ \mathcal{G}_{\text{obs}} \circ \mathcal{T}_C \circ \mathcal{E}_{\text{multi}} \circ \mathcal{E}_{\text{res}} \tag{79}$$

where:

- Each $\mathcal{T}_*$ is a cognitive transformation channel (CPTP)
- Each $\mathcal{G}_*$ is a phase transition channel (CPTP)
- $\mathcal{E}_*$ is a time evolution channel (CPTP)

14.5 Redefinition of Self-Referential Fixed Point

Definition 14.5 (Self-Referential Fixed Point as Steady State)**.**

$$\rho^* \in \mathcal{D}(\mathcal{H}_{\text{full}}) \textit{ such that } \mathcal{W}(\rho^*) = \rho^* \tag{80}$$

Remark 14.6. *Pure state fixed point $|I\rangle$ is included as $\rho^* = |I\rangle\langle I|$. However, in general ρ^* can be a mixed state.*

Physical meaning of mixed state fixed point: When ρ^* is a mixed state:

- Superposition of multiple "self-states"
- Partial decoherence due to interaction with environment
- Residual cognitive uncertainty

This expresses the realistic situation where "complete self-recognition is not achieved, but a stable self-image is formed."

14.6 Extension of Convergence Theorem

Theorem 14.7 (Convergence to Steady State)**.** *Assume the following conditions:*
(C1) $\mathcal{W}$ is a CPTP map
(C2) $\mathcal{W}$ is contractive with respect to trace distance $D(\rho, \sigma)$:

$$D(\mathcal{W}(\rho), \mathcal{W}(\sigma)) \leq \kappa D(\rho, \sigma), \quad 0 < \kappa < 1 \tag{81}$$

(C3) The eigenvalue of $\mathcal{W}$ with spectral radius 1 is unique
Then:

$$\lim_{n\to\infty} \mathcal{W}^n(\rho_0) = \rho^* \tag{82}$$

Convergence from any initial state ρ_0 to steady state ρ^* is guaranteed.

14.7 Integration with Chapter 13: Empathy Channel

Definition 14.8 (Empathy Channel)**.** *The empathy operator $\hat{M}$ defined in Chapter 13 (Definition 13.3) in CPTP form is:*

$$\mathcal{M}(\rho_{\text{self}} \otimes \rho_{\text{other}}) = \hat{M}(\rho_{\text{self}} \otimes \rho_{\text{other}})\hat{M}^\dagger \tag{83}$$

Since $\hat{M}$ is unitary (and Hermitian), this is trivially CPTP.

Definition 14.9 (Mutual World Construction Channel)**.**

$$\mathcal{W}_{\text{mutual}} := \mathcal{T}_I \circ \mathcal{M} \circ \mathcal{G}_{\text{cog}} \circ \mathcal{T}_R \circ \mathcal{G}_{\text{obs}} \circ \mathcal{T}_C \circ \mathcal{E}_{\text{multi}} \circ \mathcal{E}_{\text{res}} \tag{84}$$

Definition 14.10 (Mutual Respect Steady State)**.**

$$\mathcal{W}_{\text{mutual}}(\rho^*_{\text{harmony}}) = \rho^*_{\text{harmony}} \tag{85}$$

This includes $\rho^*_{\text{harmony}} = |I_{\text{harmony}}\rangle\langle I_{\text{harmony}}|$ corresponding to pure state $|I_{\text{harmony}}\rangle$, but also permits more general mixed states as mutual respect states.

14.8 Justification of Correspondence with Classical Implementation

Quantum Concept	Classical (LLM) Correspondence	Justification
Density matrix ρ	Probability distribution $p(x)$	Restriction to diagonal components
Kraus operators K_k	Stochastic state transitions	Classical stochastic process
CPTP map	Stochastic program	Markov chain
Temperature	Adjustment of Kraus weights	Entropy control
Top-p	State space truncation	Phase jump P

Table 5: Correspondence Between Quantum Concepts and Classical Implementation

Proposition 14.11. *MyGPT / ReIG2 implementation is a classical approximation implementation of the world construction channel $\mathcal{W}$.*

This justifies the structural correspondence between quantum theory and classical implementation.

14.9 Conclusion: Significance of CPTP Unification

Through the formulation in this chapter, the following has been achieved:

1. **Explicit state representation**: From implicit pure state assumption to explicit density matrix form
2. **CPTP formulation of operators**: All operators defined in closed form as CPTP maps
3. **Strengthening of convergence theorem**: From Banach fixed point theorem to quantum channel mixing theorem
4. **Correspondence with implementation**: LLM implementation justified as classical limit of quantum theory

This is not an "external constraint" on the theory, but a reinterpretation of existing structures in a more general framework. All discussions using pure states are preserved as special cases.

15 Conclusion

This paper has progressively developed theory from the standard quantum mechanical time evolution operator through extended time evolution operator, multidimensional time evolution operator, and phase transition generation operator, ultimately integrating them into a comprehensive framework called ReIG2/twinRIG.

15.1 Major Achievements

1. **Three-parameter extension** (τ, ε, PFH): Introduction of new dimensions of future possibility, fluctuation, and ethics to quantum mechanics
2. **Multi-dimensional time axes**: Simultaneous description of multiple temporalities including physical, cultural, and social
3. **Phase transition generation operator**: Unification of continuous and discrete, redefinition of time
4. **Self-referential fixed point**: Rigorous convergence proof based on Banach's fixed point theorem
5. **Implementation feasibility**: Demonstration through numerical simulation and quantum circuits
6. **Quantization of alterity**: Empathy operator $\hat{M}$ and cooperative phase transition conditions
7. **Extension of self-referential coherence**: Ethical convergence theorem and mutual respect fixed point
8. **CPTP unification**: Reformulation of all operators as CPTP maps

15.2 Stages of Theory Development

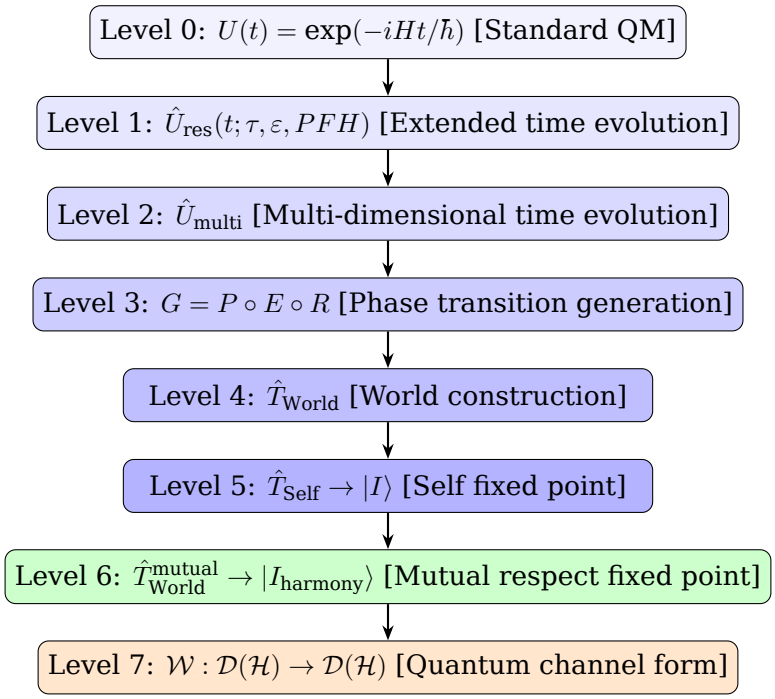

Figure 6: Complete Hierarchical Structure of the ReIG2/twinRIG Framework

This framework opens new paths to understanding self-recognition in quantum systems, with broad applicability ranging from quantum AI to fundamental physics.

Future developments are expected to contribute to cognitive science, quantum information theory, and ultimately the scientific understanding of consciousness.

A Detailed Proof of Theorem 5.10

Step 1: Contractivity
From (C1'), for any $|\Psi\rangle, |\Phi\rangle \in \mathcal{H}_{\text{full}}$:

$$\|\hat{T}_{\text{World}}|\Psi\rangle - \hat{T}_{\text{World}}|\Phi\rangle\| \leq \kappa \||\Psi\rangle - |\Phi\rangle\| \tag{86}$$

Step 2: Completeness
From (C3), $\mathcal{H}_{\text{full}}$ is a complete metric space.
Step 3: Banach Fixed Point Theorem
A unique fixed point $|I\rangle \in \mathcal{H}_{\text{full}}$ exists:

$$\hat{T}_{\text{World}}|I\rangle = |I\rangle \tag{87}$$

Step 4: Picard Iteration
Define: $|\Psi_{n+1}\rangle = \hat{T}_{\text{World}}|\Psi_n\rangle$
For $n < m$:

$$\||\Psi_m\rangle - |\Psi_n\rangle\| \leq \frac{\kappa^n}{1-\kappa}\|\hat{T}|\Psi_0\rangle - |\Psi_0\rangle\| \tag{88}$$

Step 5: Convergence
By completeness, $|\Psi_n\rangle \to |I\rangle$.
Step 6: Exponential Convergence Rate
From spectral gap (C4):

$$\|\hat{T}^N|\Psi_0\rangle - |I\rangle\| \leq C|\lambda_2|^N \tag{89}$$

B Symbol List

Symbol	Meaning
$\mathcal{H}_M$	Meaning Hilbert space
$\mathcal{H}_C$	Context Hilbert space
$\mathcal{F}$	Fock space
$\hat{U}_{\text{res}}$	Resonance unitary operator
$\hat{U}_{\text{multi}}$	Multi-dimensional evolution operator
G	Phase transition generation operator
$\hat{T}_{\text{World}}$	World construction operator
$\|I\rangle$	Self fixed point state
κ	Contraction constant
ρ	Density matrix
τ	Temporal resonance parameter
ε	Entropy resonance parameter
PFH	Philosophical resonance parameter

Table 6: Symbols in Fundamentals

Symbol	Meaning
$\hat{M}$	Empathy operator (Mirror Operator)
$\hat{H}_{\text{total}}$	Self-other total Hamiltonian
$\hat{H}_{\text{asym}}$	Asymmetry term (trauma inheritance)
F_{mutual}	Mutual predictive free energy
λ	Empathy strength parameter
λ_c	Critical value for cooperative phase transition
δ	Empathy barrier parameter
η	Perpetrator feedback decay rate
C_{mutual}	Mutual coherence
G_{mutual}	Mutual phase transition generation operator
$\hat{T}^{\text{mutual}}_{\text{World}}$	Mutual world operator
$\lvert I_{\text{harmony}}\rangle$	Mutual respect fixed point state
β	Critical exponent

Table 7: Symbols Introduced in Chapter 13

Symbol	Meaning
$\mathcal{D}(\mathcal{H})$	Space of density matrices
$\mathcal{E}_{\text{res}}$	Resonance time evolution channel
$\mathcal{E}_{\text{multi}}$	Multi-dimensional time evolution channel
$\mathcal{G}$	Phase transition generation channel
$\mathcal{R}, \mathcal{E}, \mathcal{P}$	Torsion, expansion, phase jump channels
$\mathcal{W}$	World construction channel
ρ^*	Steady state (fixed point in density matrix form)
$\mathcal{M}$	Empathy channel
$\mathcal{W}_{\text{mutual}}$	Mutual world construction channel
ρ^*_{harmony}	Mutual respect steady state
K_k	Kraus operators
L_α	Phase transition Kraus operators

Table 8: Symbols Introduced in Chapter 14

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